Exploratory Data Analysis

Return Distribution Analysis

The return distributions of the selected Turkish stocks revealed substantial differences in both risk and return characteristics. Although all assets operate within the same market environment, their sectoral exposures and business structures create distinct return behaviors.

ASELS exhibited one of the strongest long-term return profiles among the selected assets. As a major defense industry company, ASELS benefited from increased government spending, defense projects, and growing investor interest in the sector. However, the return distribution also showed higher volatility and occasional extreme observations, indicating that investor expectations and sector-specific developments can generate significant price fluctuations.

THYAO demonstrated strong return potential but also displayed the highest volatility among the selected stocks. The airline industry is highly sensitive to fuel prices, tourism activity, global economic conditions, and geopolitical developments. As a result, THYAO experienced larger positive and negative daily returns compared to most other assets.

BIMAS presented a more stable return profile. As a leading retailer operating in the consumer staples sector, the company benefits from relatively consistent consumer demand even during periods of economic uncertainty. Consequently, BIMAS exhibited lower volatility and fewer extreme return observations than cyclical stocks such as THYAO and TUPRS.

GARAN showed return characteristics closely linked to macroeconomic conditions. Banking institutions are strongly affected by monetary policy decisions, interest rates, inflation expectations, and overall economic growth. Therefore, GARAN's return distribution reflected the broader behavior of the Turkish financial system.

TUPRS generated attractive returns throughout the observation period, particularly during periods characterized by favorable energy market conditions. However, changes in oil prices, refining margins, and global energy demand contributed to elevated volatility compared to more defensive investments.

KCHOL displayed a relatively balanced return profile due to its diversified business structure. Since Koc Holding operates across multiple industries, including energy, automotive, finance, and consumer goods, the company benefits from diversification while remaining sensitive to overall economic conditions.

Overall, the return distributions were found to be non-normal, asymmetric, and heavy-tailed. This finding suggests that extreme market events occur more frequently than predicted by traditional normal distribution assumptions and highlights the importance of advanced risk management techniques.

Cumulative Return Analysis

Figure 1 presents the cumulative returns of the selected stocks and the BIST100 benchmark throughout the analysis period. cumulative return analysis revealed substantial differences in long-term investment performance. THYAO and TUPRS achieved some of the strongest cumulative returns, indicating that investors were rewarded for accepting higher levels of risk. However, these assets also experienced larger fluctuations and periods of significant drawdowns.

ASELS demonstrated strong cumulative growth driven by developments in the Turkish defense sector. Although the stock generated impressive returns during several periods, it also exhibited higher volatility compared to more stable assets.

BIMAS followed a smoother growth trajectory. While its cumulative return was not always the highest, the stock maintained relatively stable performance and experienced fewer dramatic fluctuations. This behavior reflects the defensive characteristics of the retail sector.

GARAN and KCHOL generally followed trends similar to the broader market. Their cumulative returns were influenced heavily by macroeconomic developments, monetary policy changes, and overall investor sentiment within the Turkish economy.

The comparison with BIST100 indicates that several selected stocks outperformed the benchmark over the long run. Nevertheless, market-wide movements remained an important driver of overall performance.

Correlation Analysis

To better understand relationships among assets, a correlation matrix was constructed using daily returns.

The results revealed strong positive correlations between most selected stocks and the BIST100 benchmark. This finding suggests that Turkish large-cap equities are heavily influenced by common market factors and often react similarly to changes in economic conditions.

GARAN exhibited one of the strongest relationships with BIST100. As a major banking institution, the company is highly sensitive to interest rates, inflation expectations, and monetary policy decisions. Consequently, its stock performance closely tracked broader market movements.

THYAO also displayed a strong correlation with the benchmark index. Investor sentiment, tourism demand, and economic growth expectations affect both airline performance and overall market conditions, leading to substantial co-movement.

ASELS maintained a positive correlation with the market while also exhibiting periods of independent movement driven by defense-related developments. This indicates that both market-wide and company-specific factors contribute to the stock's behavior.

BIMAS displayed comparatively lower correlations than most other assets. This result suggests potential diversification benefits, as consumer staples companies often respond differently to economic fluctuations than financial, industrial, or energy firms.

The correlation analysis confirms that systematic market risk plays a dominant role in Turkish equity markets. However, differences in correlation levels indicate that diversification opportunities still exist across sectors.

Volatility Analysis

Rolling volatility analysis was performed to evaluate how risk evolved over time.

The results revealed clear evidence of volatility clustering, a common phenomenon in financial markets where periods of high volatility are followed by additional periods of elevated volatility. This behavior indicates that market uncertainty tends to persist rather than disappear immediately.

THYAO experienced the most significant volatility spikes among all selected stocks. These fluctuations were particularly noticeable during periods of economic uncertainty and external market shocks. The findings suggest that airline stocks are highly sensitive to rapidly changing market conditions.

GARAN also exhibited elevated volatility during unstable economic periods. Since banking institutions are directly affected by changes in monetary policy and macroeconomic expectations, investor reactions often amplify price movements.

ASELS displayed moderate to high volatility throughout the sample period. While defense-sector developments supported strong growth opportunities, they also contributed to periods of heightened uncertainty.

In contrast, BIMAS maintained relatively stable volatility levels. The consistency of consumer demand helped reduce fluctuations compared to more cyclical industries.

The volatility analysis demonstrates that risk is dynamic rather than constant and that different sectors respond differently to periods of market stress.

Drawdown Analysis

Maximum drawdown analysis was conducted to evaluate downside risk and identify periods of significant losses.

THYAO experienced the largest drawdowns among the selected stocks, highlighting the substantial risks associated with high-growth opportunities. Although the stock delivered strong long-term returns, investors were required to tolerate significant temporary losses.

TUPRS also recorded notable drawdowns due to its sensitivity to energy market conditions and commodity price fluctuations.

ASELS experienced several drawdown periods associated with changes in investor expectations and market sentiment. However, the stock demonstrated the ability to recover strongly following periods of weakness.

BIMAS displayed considerably lower drawdowns than most other assets. This finding further supports its classification as a relatively defensive investment.

GARAN and KCHOL occupied an intermediate position, experiencing moderate drawdowns while remaining heavily influenced by overall market conditions.

The drawdown analysis highlights the trade-off between return potential and downside risk. Assets generating the highest long-term returns were generally associated with larger temporary losses.

Hypothesis Testing

To evaluate whether stock returns differed significantly from overall market performance, paired t-tests were conducted between each selected stock and the BIST100 benchmark.

The null hypothesis stated that the mean daily return of a stock is equal to the mean daily return of the market benchmark. The alternative hypothesis stated that a significant difference exists between the two return series.

A paired t-test was selected because stock returns and BIST100 returns are observed on the same trading days, making the observations dependent rather than independent.

The results indicate that while some stocks exhibit statistically significant differences relative to the benchmark, others move very closely with overall market performance. These findings suggest that stock-specific factors can influence performance, but market-wide dynamics remain the dominant force in determining return behavior.

Overall, the hypothesis testing results reinforce the conclusion that Turkish large-cap stocks are strongly connected to broader market conditions, while still retaining meaningful differences in risk and return characteristics across sectors.